

VLM
(10 candidates)

Avaliar fitness



OPRO
(DeepMind)

Avaliar fitness
Sort (ascending)

m iteracoes

Keep top-20 each iteration
+5 Stop criteria

$$\text{fitness} =$$

$$= 0.6 \times \text{CLIP} + 0.3 \times \frac{1}{\text{PIPS}} +$$

$$0.1 \times \frac{1}{\text{USE} + 1}$$

20 candidates sorted

como população inicial

(20)

Population
(Evolutionary)

Guardar elite

Parent selection
(Roulette wheel)

Crossover/
Mutation

LLM

GE
($\mu + \lambda$)

m Geracoes

Adjust temperature
when threshold /
Exploration first
then exploitation